

- 4 (a) State two conditions necessary for the formation of stationary waves.

The two waves must have the same frequency, same wavelength and same speed (state at least 2 of the 3).
The two waves must meet in opposite direction.

[2]

- (b) With the aid of a labelled diagram, describe a laboratory experiment to demonstrate stationary waves in an air column.

Detailed diagram of air column partially submerged in water-filled glass cylinder/
loudspeaker facing closed tube.

Use of tuning fork and varying unsubmerged length of air column/ use of varying
frequency sound source projected on closed tube.

Observing successive loud and soft sound as unsubmerged length is varied/ use of
microphone with c.r.o. to observe high and low amplitude within the closed tube.

[3]

- (c) Fig. 4.1 shows an arrangement used to determine the speed of microwaves in the laboratory.

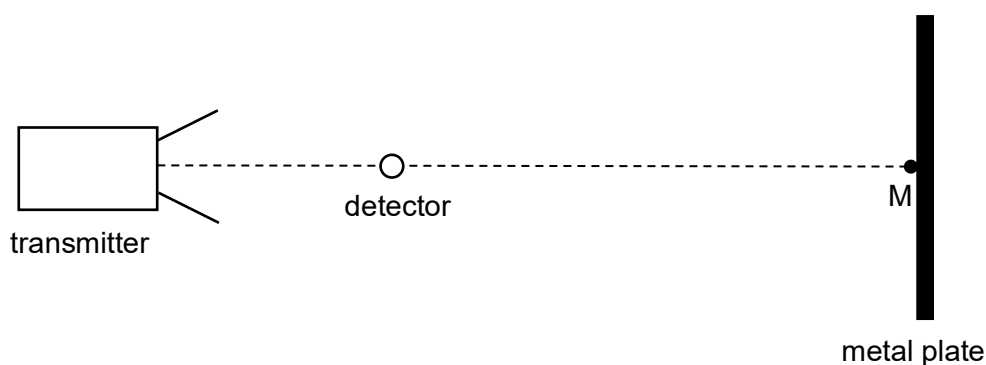


Fig. 4.1

A stationary wave is formed between the transmitter and the metal plate. As the detector is moved along the line between the transmitter and the point M, it registers a series of high intensity signals. The positions of these high intensity signals along an axis are marked by crosses on the x axis of the graph in Fig. 4.2.

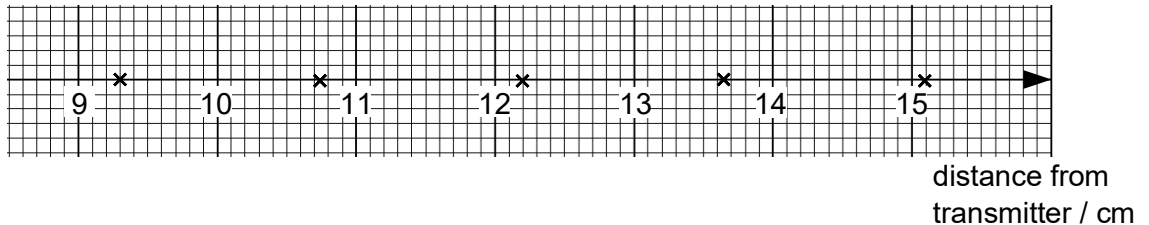


Fig. 4.2

The transmitter emits microwaves of frequency 1.0×10^{10} Hz. Use Fig. 4.2 to deduce an experimental value for the speed of the microwaves.

From Fig. 4.2, wavelength = $(15.10 - 9.30) / 2 = 2.90$ cm

Speed = $f\lambda = 1.0 \times 10^{10} (2.90 \times 10^{-2}) = 2.9 \times 10^8$ m s⁻¹

speed = m s⁻¹ [2]

[Total: 7]